CHINMOY SARDAR

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RESEARCH INTERESTS

Water Security and Quality; Climate Resilient and Decentralised Water Supply; Drinking Water Treatment and Disinfection; Water Scarcity and Climate Adaptation.

B.Sc. in Civil Engineering (Environmental major) from BUET with an undergraduate thesis built on field sampling, laboratory microbiology and statistical analysis of stored rainwater in coastal Bangladesh.

EDUCATION

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh Feb 2020 – Mar 2025
B.Sc. in Civil Engineering (Environmental Engineering specialisation) | CGPA 3.30 / 4.00

- **Grade A+ (4.00)** in undergraduate thesis, capstone project, Environmental Engineering Laboratory and Socio-economic Aspects of Development Projects.
- **Selected coursework:** Environmental Engineering I & II; Environmental Engineering Laboratory; Solid and Hazardous Waste Management; Environmental and Sustainable Management; Fluid Mechanics; Hydrology, Irrigation and Flood Management.

RESEARCH EXPERIENCE

Undergraduate Thesis, BUET Department of Civil Engineering 2024 – 2025
Variation of Rainwater Quality During Storage in Non-Rainy Seasons: A Case Study in a Coastal Area of Bangladesh
Supervisor: Prof. Dr. Md. Abdul Jalil | Conducted with one thesis partner; responsible for field sampling, laboratory analysis and statistical analysis.

- **Question:** Does stored household rainwater stay safe to drink through the dry season in Koyra, Khulna, where saline groundwater makes rainwater the main drinking source?
- **Design and methods:** Followed ten household rainwater tanks (five plastic and five brick; roof or polythene catchments) from the start of storage through four sampling rounds (August–December 2024; 40 samples). Measured pH, turbidity and total and faecal coliform (membrane filtration) at the BUET Environmental Engineering Laboratory.
- **Finding 1 (microbial safety declines with storage):** No coliforms at the start of storage; every stored sample exceeded the WHO guideline (0 CFU/100 mL) from October. In November total coliform reached **67.2 ± 8.1 CFU/100 mL in brick tanks** versus 45.6 ± 5.9 in plastic tanks; brick tanks had **2.2 times more total and 4.0 times more faecal coliform**.
- **Finding 2 (clear water is not safe water):** Turbidity in plastic tanks halved every 31–37 days and reached 0.5 NTU by December, yet every sample at or below 1 NTU still contained coliforms.
- **Analysis:** Repeated measures ANOVA, linear mixed models with tank random effects, and Welch and Mann–Whitney tests with Holm correction (Python). Implications: plastic tanks and household disinfection.
- **Household survey ($n = 100$)** of rainwater system practices in neighbouring villages: only 35% of households considered their stored rainwater safe to drink, and brick tank users reported microbial problems far more often than plastic tank users (55% vs. 20%).

PROJECTS

Capstone Project: Environmental Impact Assessment for a multistorey commercial complex, Dhaka 2024 – 2025
Deputy Team Leader of a team of 7 members over two terms | Grade A+

- Prepared the Environmental Impact Assessment, including water demand, wastewater generation, environmental safeguards and regulatory compliance; developed mitigation measures informed by a field survey.

Workshops and Outreach

Instructor, *In Search of Meaning* (photography workshop), University of Asia Pacific — Oct 2025
Instructor, *LOOP: An Introductory Photography Workshop*, BUET — Jan 2024
Volunteer, USAID Climate Resilient Ecosystems and Livelihoods (CREL) school initiative — 2015 – 2017

Honors and Awards

Dutch-Bangla Bank Merit Scholarship (HSC 2019); renewed annually through undergraduate studies — 2019 – 2025
Government Merit Scholarship (Higher Secondary Certificate) — 2019
Shortlisted (top 416) in the Royal Photographic Society 165th International Photography Exhibition — 2023

Leadership and Service

Vice President, BUET Photographic Society; led an interuniversity exhibition — 2024 – 2025
Team Leader, surveying camp at BUET — 2024

Skills

Laboratory and field: pH, turbidity and membrane filtration coliform analysis; household water sampling; longitudinal field monitoring; household surveys.
Data analysis: Python (pandas, SciPy, statsmodels) for t-tests, ANOVA, linear mixed models and multiple comparison correction; R (basic); GIS for water applications (introductory).
Domain: Environmental impact assessment; water supply and wastewater system basics; hydrology.

Standardised Tests

GRE General Test (Sep 2026): Quantitative 162, Verbal 147, Analytical Writing 3.0
IELTS Academic (Nov 2025): Overall 7.0

References

Prof. Dr. Md. Abdul Jalil, Professor, Department of Civil Engineering, BUET (thesis supervisor), majalil@ce.buet.ac.bd
Mushtari Maliha, Assistant Professor (On Study Leave), Department of Water Resources Engineering, BUET; PhD Candidate, Department of Biological Systems Engineering, Virginia Tech, mushtari@vt.edu
Shantanu Paul, Assistant Professor (On Study Leave), Department of Civil Engineering, BUET, shantanu47@ce.buet.ac.bd